

Transformer Exercise (IceCube 2D)

5/18/2025

3 Möjliga Poäng

Försök 1



Pågår

NÄSTA: Lämna in uppgift



Lägg till kommentar

Obegränsat antal försök tillåts

5/19/2025

Information

In this exercise, you will implement a Transformer to reconstruct the position of the neutrino interaction from the measured Cherenkov light.

Please see [IceCube dataset \(https://uppsala.instructure.com/courses/102453/pages/icecube-dataset\)](https://uppsala.instructure.com/courses/102453/pages/icecube-dataset) and [gnn_neutrino_dataset_description.pdf \(https://uppsala.instructure.com/courses/102453/pages/gnn_neutrino_dataset_description.pdf\)](https://uppsala.instructure.com/courses/102453/pages/gnn_neutrino_dataset_description.pdf)

for a description of the dataset. Every event consists of a list of observed Cherenkov light hits.

Please implement a Transformer with at least two heads and at least two layers. Hint: you can use these PyTorch classes: <https://pytorch.org/docs/stable/generated/torch.nn.TransformerEncoder.html>.

Because the implementation of a Transformer with different length sequences requires rather technical padding and masking, we provide a rather extensive

Evaluate the performance by calculating relevant metrics and visualizing the results in appropriate plots.

Remember always to include:

- A **written summary** (0.5–1 A4 page) covering (submitted either as PDF or directly as text):
 - What you did and how
 - What results you obtained
 - What challenges you encountered and what could be improved
- A **PDF (or similar format)** with all **result plots**, each with a short explanation
- Your **code**, preferably as a link (e.g., GitHub, Google Colab, etc.) so we can view it easily.

For those of you who want to test your skills on a realistic dataset, you can check out this Kaggle challenge of the IceCube Neutrino Observatory: <https://www.kaggle.com/icecube-neutrino>



Kom ihåg att den här inlämningen räknas för alla i din PhD exercise group grupp.

Välj en inlämningstyp

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Text



Ladda upp



Canvas Studio



Mer



Skicka uppgift

(<https://uppsala.instructure.com/courses/102453/modules/items/1370712>)